STEM Programs - Post Assessment Analysis Report

Introduction

This post-assessment analysis builds upon the earlier *STEM Data Intern v4* report, which focused on program tagging and pre-assessment score analysis. This final report evaluates student learning outcomes based on post-assessment scores across several STEM programs: Math, Virtual Reality (VR), Circuits, Hands-On Modules (HOM), CAD, Minecraft, Microbit, and Scratch.

The central research question guiding this work is:

- *Did students who participated in each STEM program show statistically significant improvements in their post-assessment scores compared to pre-assessment performance?*

Because the dataset lacks unique identifiers, we cannot pair individual students' pre- and post-assessments. Instead, we use non-parametric group-level statistical tests to estimate potential learning gains.

Sample Sizes and Data Cleaning

Dataset	Count
Pre-assessment	3,196
Post-assessment	2,265
Math (Pre)	2,893
Math (Post)	2,062
Total Math	4,955

Only records with non-NA values for *score_per* (assessment score percentage ranging from 0-10) were included in both pre- and post-assessment data.

The Math dataset comprises approximately 91% of the total pre- and post-data and was the most consistently delivered program.

Methodology

Objective

To evaluate program effectiveness through pre-/post-score comparisons for each STEM program and by student grade level.

Statistical Approach

- **Data Cleaning**:
  - Excluded any entries with NA in *score_per*.
  - Program tagging used keyword character matching (e.g., “VR+”, “Scratch”).
- **Normality Testing**:
  - Shapiro-Wilk tests showed all score distributions were non-normal ($p < 0.05$).
- **Primary Test**:
  - Wilcoxon Rank-Sum Test for independent, non-parametric group comparisons.
- **Robustness Check**:
  - Welch’s t-test (for Math) to confirm Wilcoxon results.

Math Program - Summary Statistics

Group	Mean Score	N
math_pre	0.5714	2,893
math_post	0.6457	2,062

- **Welch’s t-test**: $t = 11.888$, $p < 2.2e\text{-}16$
- **Wilcoxon test**: $W = 2{,}396{,}886$, $p < 2.2e\text{-}16$

Conclusion: There is a statistically significant improvement in post-assessment Math scores.

Program Participation Counts

Program tagging was performed using character matching (e.g., “VR+”, “Scratch”), which may result in some over- or under-matching.

Program	Pre Count	Post Count
VR	1,977	1,537
Circuits	1,459	831
HOM	1,523	874
CAD	430	250
Minecraft	212	128
Microbit	487	201
Scratch	71	74

Note: Tagging used character matching, which may lead to over- or under-identification. For example, a project with strings like “coordinates” would be tagged as VR and CAD, although they may have only been part of VR or CAD respectively.

Program-Level Score Distributions

Virtual Reality (VR)

Score	Pre	Post
0	7	4
0.1	20	9
0.2	66	36

0.3	157	102
0.4	270	140
0.5	359	229
0.6	359	250
0.7	268	210
0.8	239	198
0.9	177	245
1.0	55	114

Circuits

Score	**Pre**	**Post**
0	10	5
0.1	27	8
0.2	75	32
0.3	124	40
0.4	205	76
0.5	303	156
0.6	225	140
0.7	218	105

0.8	157	115
0.9	83	109
1.0	32	45

HOM

Score	**Pre**	**Post**
0	10	4
0.1	28	10
0.2	78	32
0.3	130	42
0.4	210	86
0.5	318	149
0.6	238	150
0.7	224	109
0.8	168	123
0.9	86	123
1.0	33	46

CAD

Score	**Pre**	**Post**

0	5	-
0.1	16	-
0.2	37	6
0.3	38	5
0.4	68	15
0.5	95	30
0.6	56	38
0.7	62	40
0.8	35	55
0.9	15	45
1.0	3	16

Minecraft

Score	**Pre**	**Post**
0	4	4
0.1	3	2
0.2	8	7
0.3	14	5
0.4	27	18

0.5	42	17
0.6	42	28
0.7	26	11
0.8	25	15
0.9	14	13
1.0	7	8

Microbit

Score	**Pre**	**Post**
0	1	-
0.1	9	3
0.2	21	3
0.3	59	20
0.4	67	16
0.5	119	42
0.6	66	34
0.7	77	30
0.8	29	23
0.9	32	20

1.0	7	10

Scratch

Score	Pre	Post
0.1	2	2
0.2	5	-
0.3	6	4
0.4	15	11
0.5	10	9
0.6	11	12
0.7	7	10
0.8	15	13
0.9	-	11
1.0	-	2

Program-Level Findings: Wilcoxon Test Results

Program	p-value	Interpretation
VR	< 2.2e-16	Significant improvement
Circuits	1.30e-14	Significant improvement

HOM	< 2.2e-16	Significant improvement
CAD	< 2.2e-16	Significant improvement
Minecraft	0.4574	No significant change
Microbit	5.78e-05	Significant improvement
Scratch	0.0034	Significant improvement

Interpretation:

- The **most substantial effects** were observed in **VR, HOM, and CAD**, all with p-values far below 0.001.
- **Minecraft** stands out as the **only program that has not shown measurable gains**.
- Despite a small sample size, **Scratch and Microbit** still showed statistically significant score improvements.

<u>Math Score Gains by Grade</u>

Wilcoxon Results by Grade - Interpretation

Grade	Pre N	Post N	p-value	Significant?	Notes
K	4	0	-	No	No post-test data
TK	2	0	-	No	No post-test data
1	26	0	-	No	No post-test data
2	28	0	-	No	No post-test data
3	123	48	0.3465	No	Small post N

4	310	232	0.1125	No	Moderate sample
5	367	295	0.00071	Yes	Strong gains
6	542	392	1.8e-12	Yes	Very strong evidence
7	618	469	2.7e-09	Yes	Strongest N
8	495	350	7.3e-09	Yes	Large effect
9	133	98	1.2e-05	Yes	Strong despite small N
10	85	49	0.0496	Borderline	Just under 0.05
11	63	62	0.00025	Yes	Strong result
12	78	67	0.0148	Yes	Meaningful gains

- **Grades 5-12 consistently showed improvement**, with **Grades 6-9** yielding the most robust results.
- Grades 3 and 4 did not exhibit significant score changes, possibly due to developmental factors or differences in program delivery.
- Grades K-2 were excluded from testing due to the lack of post-assessment data, despite some participation in the pre-assessment.

<u>Limitations</u>

Program Tagging

- Character matching (e.g., "VR+") is used to tag programs
- Risk of over-tagging (false positives) or missing students (false negatives)

- No manual or context validation

Sample Size

- Small N for some programs (e.g., Scratch, Minecraft, CAD) could lead to unstable trends.
- Some grades were excluded due to missing post-assessment data.
- Sample sizes varied widely by grade, which may affect statistical power.
- For younger grades, the assessment content or delivery format may not have been fully aligned with the students' skill level, which can affect score consistency.

Lack of Matched Data

- Cannot track individual student growth across pre-/post.
- Group-level tests may mask within-student variability.

Score Distribution

- Possible ceiling or floor effects skewing scores

Instructional Fidelity

- No data on implementation quality, instructional time, or teaching differences

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Summary

The analysis indicates that several STEM programs likely contributed to meaningful learning gains:

- **Scores increased significantly overall**, particularly for students in grades **5 through 12**.
- Substantial gains were observed in **VR**, **Circuits**, **Hands-On Modules**, and **CAD** participants.
- **Microbit and Scratch** also saw measurable improvements.
- **Minecraft showed no significant change**, possibly due to its open-ended, non-assessment-aligned nature.

Despite limitations, these findings suggest that the **majority of program offerings had a positive impact on student learning**, with clear implications for scaling and future targeting.

Implications

- Focus expansion efforts on programs with the **strongest learning outcomes**, such as VR, HOM, and CAD.
- Consider revising or aligning the **Minecraft curriculum** more closely with assessment goals.
- Improve **data tagging protocols** and ideally implement **unique student identifiers** for longitudinal tracking.
- Further investigate outcomes by **demographics**, **session hours**, or **instructor differences** to refine insights.

Program-by-Grade Analysis of Score Gains

Methodology

To better understand how each STEM program may have affected students at different grade levels, Wilcoxon rank-sum tests were conducted separately for each program, comparing **pre- and post-assessment scores within each grade**. Only records with valid *score_per* values were included. Grades with fewer than 10 students in either pre- or post-groups were excluded from hypothesis testing due to insufficient sample size.

Summary Table: Significant Score Gains by Program and Grade

Program	Grade	Pre N	Post N	p-value	Statistically Significant?
Virtual Reality	5	280	238	0.01206	Yes
	6	370	295	1.857e-08	Yes
	7	469	341	6.972e-09	Yes
	8	343	244	9.022e-05	Yes
	9	77	76	0.01524	Yes

Circuits	3	48	31	0.03281	Yes (small N caveat)
	6	246	144	4.376e-07	Yes
	7	248	142	0.00044	Yes
	8	212	115	2.516e-05	Yes
	9	74	30	0.01047	Yes
HOM	3	48	31	0.03281	Yes (small N caveat)
	6	263	167	7.002e-06	Yes
	7	263	160	2.537e-05	Yes
	8	229	126	1.935e-05	Yes
	9	74	29	0.007984	Yes
CAD	6	99	43	8.236e-07	Yes
	7	80	40	0.000364	Yes
	8	92	43	4.668e-07	Yes
	9	27	19	0.000651	Yes (small N)
	10	20	16	0.00992	Yes (small N)
	11	38	39	0.000382	Yes
	12	41	41	2.372e-05	Yes

Microbit	5	95	45	0.008558	Yes
	6	92	33	0.02505	Yes
	11	15	16	0.00118	Yes (small N)
	12	16	21	0.0002882	Yes
Scratch	5	11	21	0.09449	No (Marginal; small N)
Minecraft	—	—	—	All $p > 0.1$	No significant changes

Key Patterns & Statistical Highlights

Virtual Reality (VR)

- Grades 5-9 all showed significant gains ($p < 0.05$), with Grades 6-8 having the strongest sample sizes and most compelling results:
    - Grade 6: N = 370 (pre), 295 (post); p = 1.86e-08
    - Grade 7: N = 469 (pre), 341 (post); p = 6.97e-09
    - Grade 8: N = 343 (pre), 244 (post); p = 9.02e-05
- Interpretation: This suggests that the VR curriculum is particularly effective in upper elementary and middle school years.

Circuits & HOM (Hands-On Modules)

- Both showed nearly identical improvement patterns across Grades 6-9, with p-values consistently < 0.001.
    - Grade 6 (Circuits): p = 4.38e-07
    - Grade 6 (HOM): p = 7.00e-06
    - Even in Grade 3, both groups showed significant gains (p = 0.0328); however, the sample sizes were small (N = 48 and 31), so the findings should be interpreted with caution.

CAD (Computer-Aided Design)

- CAD was the only program to demonstrate significant gains across all secondary grades tested (Grades 6-12):
    - Grade 6-9: p-values ranged from 8.24e-07 to 0.00065
    - Grades 10-12: All showed p-values < 0.01 despite smaller sample sizes
- Interpretation: CAD yields broad learning gains across all grade levels, indicating strong curricular alignment.

Microbit

- Gains were significant in:
    - Grade 5: N = 95 (pre), 45 (post); p = 0.0086
    - Grade 6: N = 92 (pre), 33 (post); p = 0.0251
    - Grades 11-12: N = 15-21 per group; p-values < 0.001
- Note: High school Microbit N sizes were very small, so results should be treated as exploratory.

Scratch

- No grade group reached significance. Grade 5 had a p-value of 0.0945 (borderline), but a small sample size limits interpretability.

Minecraft

- No significant score changes were detected in any grade.

Key Findings

- **Grades 6-9 showed the most consistent program-driven gains**, particularly in VR, Circuits, HOM, and CAD.
- **CAD had a broad impact across all tested secondary grades (6-12), with statistically significant gains observed in all tested grades**.
- **Microbit saw gains in Grades 5, 6, 11, and 12**, although sample sizes in high school were small.

- **Scratch and Minecraft** had limited sample sizes, especially in younger grades, and **did not show significant gains** in most groups.
- **Virtual Reality (VR)** showed substantial gains in Grades 5-9, particularly in Grades 6-8, aligning with trends observed in the general math score analysis.

Additional Limitations (Specific to Program-by-Grade)

- **Sparse or missing data** for Grades K-2 and Grade 3 in most programs; excluded from testing due to low N.
- Several **p-values were computed with ties**, which can affect the reliability of the Wilcoxon test; this occurred in small sample sizes (e.g., Grade 12 in Microbit).
- This level of disaggregation reduces **statistical power**, increasing the risk of false negatives or Type II errors in smaller groups.
- **Non-random assignment** and inconsistent program exposure by grade or site are likely; without implementation data, it's unclear whether observed gains are due to curriculum exposure or confounding variables.
- **Low Power in High School Grades**:
    - Many program-grade combinations in Grades 10-12 had small sample sizes ($N < 25$), which reduced the stability of the results.

Interpretation & Implications

- **Grades 6-8** are the strongest zone of program impact, especially across VR, CAD, Circuits, and HOM.
- **CAD** is unique in its broad impact across middle and high school grades (6-12), suggesting high versatility.
- **Microbit** showed late-grade gains (11-12), possibly due to greater cognitive readiness or project alignment.
- **Scratch** and **Minecraft** require either curricular revision or alternate assessments better suited to their formats.